# A LEAST-SQUARES LINEAR LENS

GIORGIO YOUNG

Abstract. In this note, we examine several global linearization techniques and examine some qualitative attributes of the slope variable $\mathcal{J}$ under each method in comparison to the $J$-lens introduced in [1]. If the preliminary results are positive, we will construct a corresponding $\mathcal{J}$-space analogously to [1], and examine how its attributes compare. We will also examine whether additional affine or quadratic terms in the approximation improve the workspace representation in some sense.

## 1. Introduction

We set some notation and review the $J$-space construction introduced in [1]. Consider a causal autoregressive transformer with $L$ layers. Given an input sequence $x$ consisting of $n = n(x) \in \mathbb{N}$ tokens $x = (t_1, \dots, t_n)$, we denote by

$$h_\ell^{(i)} = h_\ell^{(i)}(t_1, \dots, t_i) \in \mathbb{R}^d$$

the internal representation at a layer $1 \le \ell \le L$, where $d = d_{model}$ is the dimension of the residual stream, or the length of the vector embeddings. In [1], the $J$-lens is defined to be the Jacobian in the $\ell$'th layer internal representations averaged over inputs and admissible tokens:

$$J_\ell = \mathbb{E}_{x,(i,j)\in P}\left[\frac{\partial h_L^{(i)}(x)}{\partial h_\ell^{(j)}(x)}\right] \in \mathbb{R}^{d\times d},$$

where $P = P_x$ is an admissible set of tokens:

$$P = \{(i,j) : 17 \le j \le i \le n(x) - 1\}$$

where the lower bound is empirically motivated, and the condition $i \le j$ is a consequence of causality (nb, we are indexing from 1 here). In other words, a local first order approximation, followed by a an average over prompts and admissible tokens.

It is natural to ask about a more global "linearization," i.e. a least squares approximation, that has the feature of being a global approximant in inputs and tokens to start, and this will be our focus in this note. We will attempt to construct and test this new “global" $J$-lens, from which the remaining $J$-space constructions will follow similarly. We note here that the set-up we describe below has already appeared in [2]. The authors relate this formulation to the original $J$-lens construction, showing via Stein's identity that the optimal least-squares slope coincides with the expected Jacobian of the corresponding readout map when its input distribution is Gaussian and under suitable regularity assumptions. The explorations below are novel, to the best of our knowledge. In addition to being natural and interesting, this approach would be numerically cheaper than the derivative computation involved in producing the J-lens. If our results are negative, there may be something fairly subtle in the J-lens paper, and probably more questions to be answered.

**Question 1.1.** Can we globally linearize to start via a least squares procedure and recover the attributes of the $J$-lens? Can it be improved in accuracy at low computation cost by adding quadratic and/or affine term?

We note some advantages to a positive resolution include computational efficiency, and the modification is mathematically natural, with these extensions similarly not too expensive computationally. Indeed, we will only need forward passes and no derivatives should be computed. There are several inequivalent potential ways to more globally approximate, and we will test each empirically on a small parameter model.

We set

$$\begin{aligned} V_x &= \{17, \dots, n(x) - 1\}, \\ P_x &= \{(i, j) : 17 \le j \le i \le n(x) - 1\}. \end{aligned}$$

(1) Pooled: The most naive approach is the pooled causal-pair regression: this minimizes distance over both inputs and tokens, privileging neither:

$$(A_\ell^{\text{pooled}}, b_\ell^{\text{pooled}}) \in \arg \min_{\substack{A \in \mathbb{R}^{d\times d} \\ b \in \mathbb{R}^d}} \mathbb{E}_{x,(i,j)\in P_x} \left[ \left\| h_L^{(i)}(x) - A h_\ell^{(j)}(x) - b \right\|^2 \right], \tag{1.1}$$

and set

$$\mathcal{J}_\ell^{\text{pooled}} = A_\ell^{\text{pooled}}.$$

(2) Same: Same position regression: restricts to same token matching downstream, optimizing over error at each token $j$:

$$(A_\ell^{\text{same}}, b_\ell^{\text{same}}) \in \arg \min_{\substack{A \in \mathbb{R}^{d\times d} \\ b \in \mathbb{R}^d}} \frac{1}{N} \sum_{k=1}^{N} \frac{1}{|V_{x_k}|} \sum_{j \in V_{x_k}} \left\| h_L^{(j)}(x_k) - A h_\ell^{(j)}(x_k) - b \right\|^2, \tag{1.2}$$

so that

$$\mathcal{J}_\ell^{\text{same}} = A_\ell^{\text{same}}.$$

where we select the Frobenius norm minimizing solution. As we will discuss in the empirical section, the minimizer is in fact unique in our experiment.

(3) Pair: Average over pair specific regressions, the most similar to a non-local $J$-lens: Setting

$$K_{ij} = \{k : (i, j) \in P_{x_k}\}$$

we take

$$(A_{\ell;i,j}, b_{\ell;i,j}) \in \arg \min_{\substack{A \in \mathbb{R}^{d\times d} \\ b \in \mathbb{R}^d}} \frac{1}{|K_{ij}|} \sum_{k \in K_{ij}} \left\| h_L^{(i)}(x_k) - A h_\ell^{(j)}(x_k) - b \right\|^2. \tag{1.3}$$

and define

$$\mathcal{J}_\ell^{\text{pair}} = \frac{1}{N} \sum_{k=1}^{N} \frac{1}{|P_{x_k}|} \sum_{(i,j) \in P_{x_k}} A_{\ell;i,j}.$$

## 2. Theoretical Preliminaries

We record closed form solutions for each candidate $\mathcal{J}_\ell$, although they are standard.

We may find the norm minimizing solution of (1.1) as:

$$\mathcal{J}_\ell^{\text{pooled}} = \operatorname{Cov}_{x,(i,j)\in P_x}\left(h_L^{(i)}(x), h_\ell^{(j)}(x)\right)\left[\operatorname{Cov}_{x,(i,j)\in P_x}\left(h_\ell^{(j)}(x), h_\ell^{(j)}(x)\right)\right]^\dagger$$

where $A^\dagger$ denotes the Moore-Penrose inverse of a matrix $A$.

Similarly, for (1.2), we may solve

$$\mathcal{J}_\ell^{\text{same}} = \operatorname{Cov}_{x,j\in V_x}\left(h_L^{(j)}(x), h_\ell^{(j)}(x)\right)\left[\operatorname{Cov}_{x,j\in V_x}\left(h_\ell^{(j)}(x), h_\ell^{(j)}(x)\right)\right]^\dagger$$

and (1.3) by

$$A_{\ell;i,j} = \operatorname{Cov}_{k\in K_{ij}}\left(h_L^{(i)}(x_k), h_\ell^{(j)}(x_k)\right)\left[\operatorname{Cov}_{k\in K_{ij}}\left(h_\ell^{(j)}(x_k), h_\ell^{(j)}(x_k)\right)\right]^\dagger$$

and then $\mathcal{J}_\ell^{pair}$ is the average:

$$\mathcal{J}_\ell^{\text{pair}} = \frac{1}{N}\sum_{k=1}^{N}\frac{1}{|P_{x_k}|}\sum_{(i,j)\in P_{x_k}} A_{\ell;i,j}.$$

## 3. Empirical Preliminaries

We will be using Qwen2.5-0.5B-Instruct to run our experiment. It has 24 layers and 500 million parameters, with $d = 896$ the model dimension. We will be running this locally, so a larger model will not be functional, particularly if we want to compare directly to the original $J$-lens construction, which will require backwards propagation.

We used Wikitext to fit our affine approximations, and recovered each $\mathcal{J}$ as described above, while holding out reserved Wikitext passages for evlaution. In `evaluate_lenses.py`, we compare the three methods of fitting affine lenses with the J-lens on these reserved Wikitext validation passages. We produce readouts and qualitatively compare the three affine-slope lenses with J-lens, along with ordinary logit-lens, final-layer, and constant training-mean readouts as baselines. More precisely, we compare top ten highest scoring vocabulary tokens at the midpoint of the retained interval $17 + \lfloor \frac{n-18}{2} \rfloor$, and the penultimate token positions of each passage. We apply each lens matrix to the layer-12 (the middle layer) residual, along with the model's final RMS normalization and unembedding, and inspect the ten highest-scoring vocabulary tokens. For "type" corresponding to one of our three affine lenses, pooled, paired, and same, we compute

$$\begin{aligned} s_{type}(x,j) &= W_U \ \operatorname{RMSNorm}_{\text{final}}\left(\mathcal{J}_{12}^{type} h_{12}^{(j)}(x)\right) \in \mathbb{R}^{|\mathcal{V}|}, \\ \operatorname{Readout}_{type}(x,j) &= \operatorname{Decode}(\operatorname{TopKIndices}_{10}[s_{type}(x,j)])\,. \end{aligned}$$

To illustrate the results, we record two examples. The first, our global affine lenses appear to capture semantic context better than J-lens, and in the second, worse. The similarity in the pooled outputs reflects pooled limited contextual flexibility, almost certainly downstream from the affine fit being overly constrained, as the best fit affine map with respect to a token and input combined norm. Paired instead seems to be qualitatively the most promising across contexts. We emphasize that both examples were selected after examining the results.

Context: “…He immediately ran into a problem: the South Carolina troops (militia or **`the`**”

| Lens | Top five tokens |
|---|---|
| Pooled | Canyon, Cemetery, damage, Surrey, gunfire |
| Same | Union, union, latter, unions, firing |
| Paired | cavalry, troops, militia, Protector, revolt |
| J-lens | `?\u201d`, `?"\n`, `?"`, `\uff1f\u201d`, `.\u201d` |

Table 1. Illustrative layer-12 readouts at position 127 of a 128-token validation passage. The inspected token is in bold, and tokens appear in descending score order, with leading spaces omitted. For J-lens, `\n` denotes a newline, `\u201d` a right double quotation mark, and `\uff1f` a full-width question mark.

Context: “…also named Saffar, is an extrasolar planet approximately 44 light-years away from **`the`**”

| Lens | Top five tokens |
|---|---|
| Pooled | canyon, Canyon, exposures, `\u81ea\u7136\u4fdd\u62a4`, emetery |
| Same | continent, Isle, northwest, north, island |
| Paired | northwest, coastal, north, northeast, northern |
| J-lens | Earth, orbit, planet, NASA, SpaceX |

Table 2. Illustrative layer-12 readouts at position 72 of a 128-token validation passage (`sequence_00004.pt`). The inspected token is in bold, and tokens appear in descending score order, with leading spaces omitted. The Unicode escape sequence denotes a Chinese token meaning “nature conservation”; `emetery` is a literal token fragment. Context spacing and hyphenation are simplified for readability. This example was selected after inspecting the results to illustrate a contextually relevant J-lens readout and an unrelated paired readout.

## References

1. Wes Gurnee, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, and Jack Lindsey, *Verbalizable representations form a global workspace in language models*, arXiv preprint arXiv:2607.15495 (2026).
2. Shi-Qi Yan, Kai-Xuan Ding, Chao-Hong Tan, Qian Chen, Wen Wang, Xiangang Li, and Zhen-Hua Ling, *Short horizons and sparse concepts: a mathematical view of the readout in the j-lens*, arXiv preprint arXiv:2608.25347 (2026).

*Email address*: `giorgio.young@gmail.com`